

Phase 8: Data Management & Deployment

Doctor Appointment & Consultation System (Healthcare CRM)

Objective

The main goal of this phase is to handle the data management and deployment aspects of the Salesforce project.

This phase focuses on importing, exporting, maintaining data integrity, and deploying developed components efficiently between environments.

In this project, various data management features and deployment tools have been explored, primarily focusing on Salesforce's native tools and VS Code integration.

1. Data Import Wizard

The **Data Import Wizard** is a Salesforce tool used to load data into the system.

For this project:

- Sample records (Doctors, Patients, and Appointments) were created manually using the Salesforce UI to demonstrate basic data handling.
- The Data Import Wizard was not required since bulk data upload was not part of the current scope.

2. Data Loader

The **Data Loader** is a desktop tool used for large data operations (insert, update, upsert, delete, and export).

Although not used in this phase, it remains an important Salesforce tool for:

- Exporting existing data for reporting or analysis.
- Updating bulk records, such as rescheduling appointments in bulk or updating doctor availability.

Since this project used minimal test data, manual record creation was sufficient.

3. Duplicate Rules

Duplicate management ensures that the same record isn't created multiple times for the same entity.

In this project:

- Salesforce's default duplicate management was enabled to prevent duplicate entries of patients or doctors.

This ensures data integrity and prevents redundancy in the CRM database.

4. Data Export & Backup

Data export is essential for data safety and compliance.

In this phase:

- The **Data Export Service** available in Salesforce Setup can be used to back up all data (Appointments, Prescriptions, Patients, and Doctors).
- Backups can be scheduled weekly or monthly, ensuring business continuity in case of data loss.
- Though manual export wasn't performed, it is acknowledged as an important data governance practice.

5. Change Sets

Change Sets are Salesforce's built-in method for migrating metadata between environments (for example, from Sandbox to Production).

In this project:

- **Change Sets were not used**, as the project utilized a **VS Code & SFDX**-based deployment approach, which provides more flexibility and control.

However, in enterprise scenarios, Change Sets can be used to deploy custom objects, fields, triggers, and Apex classes between connected Salesforce orgs.

6. Unmanaged vs Managed Packages

Salesforce allows packaging components for distribution:

- **Managed Packages:** Used for AppExchange applications, controlled by the publisher.
- **Unmanaged Packages:** Ideal for open projects or internal deployments.

This project followed an **Unmanaged Package** structure using **VS Code and SFDX**, as it is under development and owned by a single organization (college-based Salesforce org).

7. ANT Migration Tool

The **ANT Migration Tool** is a command-line utility for automating deployment through XML configuration files. In this project, ANT was **not used** since **VS Code & SFDX CLI** provided modern and simplified deployment capabilities.

Developers prefer SFDX for its integration with source control systems and direct deployment commands.

8. VS Code & SFDX Deployment

This project was built and deployed using **Visual Studio Code (VS Code)** integrated with **Salesforce CLI (SFDX)**.

Key Steps Implemented:

1. Salesforce project structure was created using:

```
sfdx force:project:create -n DoctorAppointmentCRM
```

2. Org was connected using:

```
sfdx auth:web:login -a DevHub
```

3. Components were deployed using:

```
sfdx force:source:deploy -p force-app
```

4. Deployment verification was done via Salesforce Org's **Setup → Apex Classes / Lightning Components**.

Advantages:

- Fast deployment and rollback.
- No dependency on manual uploads.
- Integration with Git for version control.
- Enables local development and testing before pushing to the org.

